

# **Chained Indices Unchained:**

## Structural Transformation and the Welfare Foundations of Income Growth Measurement

**Omar Licandro**

University of Leicester and BSE

**Juan Ignacio Vizcaino**

University of Nottingham

# Plan

Introduction

Stylized Facts

Economic Environment

Economic Theory of Index Numbers

Quantitative Implementation

Main Quantitative Results

Conclusion

# Motivation and Research Question

- ▶ We study how to measure **welfare-relevant growth** in economies undergoing:
  - ▶ structural transformation (ST),
  - ▶ investment-specific technical change (ISTC).
- ▶ Why does this matter? ST and ISTC lead to permanent changes in:
  - ▶ relative prices,
  - ▶ expenditure shares.

Which can generate aggregation biases in traditional quantity indices.

- ▶ In this context, we ask:
  - ▶ Do chained indices provide a welfare-based measure of real income growth?
  - ▶ Are they preferred to fixed-based equivalent variation measures?

# Our Approach

- ▶ Drivers of ST
  - ▶ Non-homothetic preferences: income effects.
  - ▶ Heterogeneous sectoral productivity growth: changes in relative prices.
- ▶ ISTC leads to
  - ▶ Permanent decline in the relative price of investment.
  - ▶ Affecting dynamic consumption-investment choices (Bellman Equation).
  - ▶ Leading to time-varying consumption-investment indifference maps.
- ▶ To deal with time-varying indifference maps, we apply the **Fisher–Shell Principle**:
  - ▶ Welfare comparisons must use a consistent preference order.
  - ▶ The natural choice is current preferences.

# Our Approach

- ▶ Alternative applications of the Fisher-Shell (FS) Principle:
  - ▶ **Fixed-Base FS Indices**
    - ▶ All past is compared to the present using current preferences and prices
    - ▶ Baqaee and Burstein (QJE, 2024)
  - ▶ **Chained FS Indices**
    - ▶ Measure infinitesimal welfare gains holding local preferences and prices fixed.
    - ▶ Then chaining these gains over time.
- ▶ Following Durán and Licandro (EJ, 2025)
  - ▶ The Chained FS quantity index coincides with the Chained Divisia index in continuous time.
  - ▶ Growth rates and expenditure shares are sufficient statistics of welfare gains.

# Main Findings

- ▶ Within the framework of Herrendorf, Rogerson, and Valentinyi (2021), featuring
  - ▶ non-homothetic preferences,
  - ▶ sector-specific productivity growth,
  - ▶ investment-specific technical change.
- ▶ We compare
  - ▶ The Chained Divisia Index.
  - ▶ A Current-Base Fisher–Shell Index (CB).
- ▶ Quantitative Findings
  - ▶ The CB Index substantially understates past welfare growth.
  - ▶ Historical growth is revised downwards as the base period moves.
- ▶ Information Requirement
  - ▶ The Divisia Index only requires expenditure shares and growth rates.
  - ▶ The CB Index additionally requires knowledge of deep model parameters.

# Plan

Introduction

Stylized Facts

Economic Environment

Economic Theory of Index Numbers

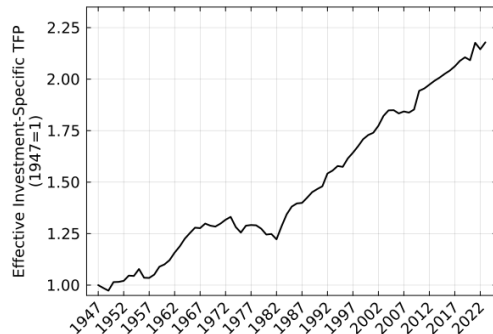
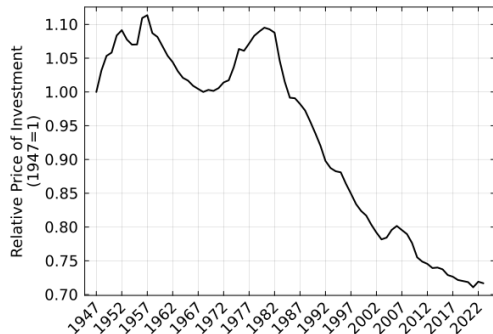
Quantitative Implementation

Main Quantitative Results

Conclusion

# ST and IBTC

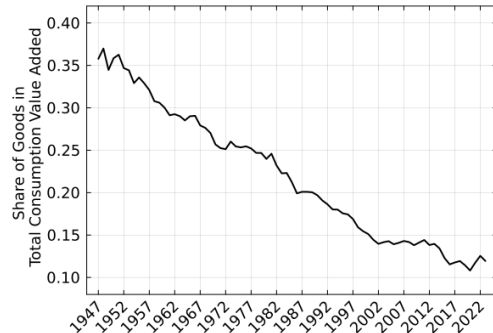
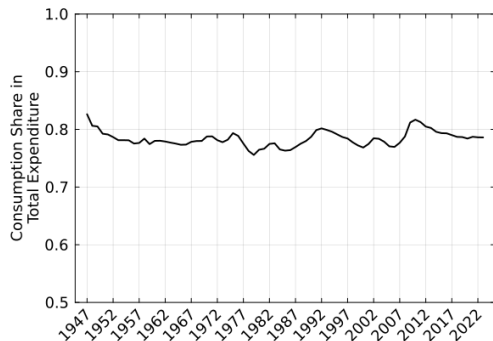
## Relative Price of Investment and Aggregate TFP



- ▶ Since 1980:
  - ▶ declining relative price of investment.
  - ▶ rising aggregate TFP, at constant rate.

# ST and IBSC

## Total and Consumption Expenditure Composition



- ▶ Consumption expenditure roughly stable as a share of GDP.
- ▶ Strong decline in the share of goods in total consumption expenditure.

# Plan

Introduction

Stylized Facts

**Economic Environment**

Economic Theory of Index Numbers

Quantitative Implementation

Main Quantitative Results

Conclusion

# Structural Transformation Model

## Sectoral Production

- ▶ Two consumption sectors: goods  $g$  and services  $s$ .
- ▶ The final good can be allocated to consumption expenditure or investment.
- ▶ Sectoral value added is produced with Cobb–Douglas technologies:

$$Y_{j,t} = A_{j,t} K_{j,t}^{\theta} L_{j,t}^{1-\theta}, \quad j \in \{g, s\}.$$

- ▶ Capital and labor are fully mobile across sectors,
- ▶ + firms face common equilibrium factor prices  $(r_t, w_t)$ :

$$\rightarrow \frac{K_{g,t}}{L_{g,t}} = \frac{K_{s,t}}{L_{s,t}}$$

# Structural Transformation Model

## Investment Technology

- ▶ Investment is produced using goods and services as intermediate inputs:

$$I_t = A_{x,t} \left[ \omega^{1/\varepsilon} X_{g,t}^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)^{1/\varepsilon} X_{s,t}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}.$$

- ▶  $X_{g,t}$  and  $X_{s,t}$  are goods and services used in investment production.
- ▶  $A_{x,t}$  captures investment-specific technical change.
- ▶ Capital evolves according to:

$$\dot{K}_t = I_t - \delta K_t.$$

# Equilibrium Prices

## Model in Investment Units

- ▶ Investment good as the numeraire.
- ▶ Equilibrium relative prices are pinned down by sectoral productivities.

$$\frac{P_{g,t}}{P_{s,t}} = \frac{A_{s,t}}{A_{g,t}}.$$

- ▶ Prices relative to investment are:

$$P_{g,t} = \frac{\mathcal{A}_t}{A_{g,t}}, \quad P_{s,t} = \frac{\mathcal{A}_t}{A_{s,t}},$$

where

$$\mathcal{A}_t = A_{x,t} \left[ \omega A_{g,t}^{\varepsilon-1} + (1 - \omega) A_{s,t}^{\varepsilon-1} \right]^{\frac{1}{\varepsilon-1}}.$$

# Aggregate Technology Representation

## Aggregation Result

- Aggregate final output, measured in units of investment, is:

$$Y_t = \underbrace{P_{g,t}C_{g,t} + P_{s,t}C_{s,t}}_{E_t = \text{consumption expenditure}} + I_t$$

- Since factor ratios are equalized across sectors:

$$Y_t = \mathcal{A}_t K_t^\theta L_t^{1-\theta}.$$

$\mathcal{A}_t$  is aggregate TFP measured in units of investment.

**Structural transformation is consistent with an aggregate production function.**

# Preferences and Structural Transformation

## PIGL Demand System

- Preferences over  $g$  and  $s$  are represented by the PIGL indirect utility function:

$$V(e_t, P_{g,t}, P_{s,t}) = \frac{1}{\chi} \left[ \left( \frac{e_t}{P_{s,t}} \right)^{\chi} - 1 \right] + \frac{\eta}{\gamma} \left[ 1 - \left( \frac{P_{g,t}}{P_{s,t}} \right)^{\gamma} \right],$$

with

$$\eta > 0, \quad 1 > \gamma > \chi > 0.$$

- Goods share in total consumption expenditure:

$$s_{g,t} = \frac{P_{g,t} c_{g,t}}{e_t} = \underbrace{\eta \left( \frac{e_t}{P_{s,t}} \right)^{-\chi}}_{\text{income effect}} \times \underbrace{\left( \frac{P_{g,t}}{P_{s,t}} \right)^{-\gamma}}_{\text{substitution effect}}$$

# Aggregate Balanced Growth Path

**ABGP:** BGP for aggregate variables + ST within Consumption and Investment

- ▶ Assume sectoral productivities grow at constant rates:

$$A_{g,t} = A_{g,0}e^{g_g t}, \quad A_{s,t} = A_{s,0}e^{g_s t}, \quad \mathcal{A}_t = \mathcal{A}_0e^{g_{\mathcal{A}} t},$$

with

$$g_{\mathcal{A}} > g_g > g_s.$$

- ▶ Along the ABGP, aggregate per-capita variables grow at:

$$g_k = g_e = g_y = \frac{g_{\mathcal{A}}}{1 - \theta} + g_h.$$

- ▶ The model features aggregate growth in investment units.
- ▶ But relative prices and expenditure shares still change over time.

# Divisia Growth Along the ABGP

## Real GDP Growth

- ▶ Along the ABGP, the Divisia growth rate of real GDP is:

$$g_t^D = s_e \underbrace{[s_{g,t}g_{c_g} + (1 - s_{g,t})g_{c_s,t}]}_{\text{Growth rate of real consumption expenditure}} + (1 - s_e)g_x.$$

- ▶  $s_e$ : share of consumption expenditure in GDP.
  - ▶  $g_x$ : growth rate of investment.
  - ▶  $s_{g,t}$ : share of goods in consumption expenditure.
- ▶ Since  $s_{g,t}$  and  $g_{c_s,t}$  change over time, and  $g_{c_g} > g_{c_s,t}$ :

$$\implies g_t^D \text{ declines along the ABGP.}$$

**The Divisia growth rate differs from the growth rate of GDP measured in investment units.**

# Plan

Introduction

Stylized Facts

Economic Environment

**Economic Theory of Index Numbers**

Quantitative Implementation

Main Quantitative Results

Conclusion

# Key Questions

- ▶ Do chained indices provide a welfare-based measure of real income growth?
- ▶ Are they preferred to fixed-based equivalent variation measures?

# Bellman Representation of Preferences

## Non-Homothetic and Time-Varying Preferences

- ▶ The Bellman representation of preferences is:

$$W_t(e, x) = \max_{\{e, x\}} V(e, P_{g,t}, P_{s,t}) + \nu_t x$$

$$\text{s.t.} \quad e + x = m_t$$

- ▶  $\nu_t$  is the marginal value of capital:  $\nu_t = v'(k_t)$ .
- ▶ The Bellman representation is:
  - ▶ Quasi-linear  $\rightarrow$  Non-Homothetic.
  - ▶ Time-varying  $\rightarrow \{P_{g,t}, P_{s,t}, \nu_t\}$  change over time.

# Bellman Representation of Preferences

## Indirect Utility and Expenditure Functions

- In equilibrium, consumption expenditure is:

$$e_t = (\nu_t P_{s,t}^\chi)^{\frac{1}{\chi-1}}.$$

- Given net income  $m$  and prices  $(P_g, P_s)$ , the indirect utility function is

$$u(m, P_g, P_s; \nu) = V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right) + \nu \left[ m - (\nu P_s^\chi)^{\frac{1}{\chi-1}} \right].$$

- The associated expenditure function is

$$f(w, P_g, P_s; \nu) = (\nu P_s^\chi)^{\frac{1}{\chi-1}} + \frac{w}{\nu} - \frac{V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right)}{\nu}.$$

- Quasi-linearity in investment makes  $m$  and  $w$  enter linearly.

# Equivalent variation Measure of Income

- ▶ Fisher-Shell Principle: Welfare comparisons require a common preference.
- ▶ Let
  - ▶  $t$ : date of the preference order,
  - ▶  $\tau$ : date whose welfare is evaluated,
  - ▶  $z$ : date of reference prices.

$$\widehat{m}_{t,z,\tau} = f(u(m_\tau, P_{g,\tau}, P_{s,\tau}; \nu_t), P_{g,z}, P_{s,z}; \nu_t).$$

- ▶  $\widehat{m}_{t,z,\tau}$ : income at prices  $z$  needed to attain the welfare of period  $\tau$ , using preferences of  $t$ .

# Proposition 3

## Equivalent Variation vs Current Income

- Under the Bellman representation,

$$\hat{m}_{t,z,\tau} = m_{\tau} + B_{t,z,\tau} \hat{e}_{t,z},$$

where

$$B_{t,z,\tau} = \left( \frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,\tau}}{\gamma} \right) \left( \frac{P_{s,z}}{P_{s,\tau}} \right)^{\frac{\chi}{1-\chi}} - \left( \frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,z}}{\gamma} \right).$$

- The EV measure equals observed income plus a price/preference correction.
- The correction depends on hypothetical consumption reallocation gains under the base preference order.

$t$  → preference date     $\tau$  → welfare comparison date     $z$  → reference price date

# Proposition 4

## Growth Rate of the Equivalent Variation Measure

- ▶ The growth rate of the EV income measure is

$$\frac{d \log \hat{m}_{t,z,\tau}}{d\tau} = \frac{m_{\tau}}{\hat{m}_{t,z,\tau}} \left( g_{\tau}^D + \text{dev}_{t,z,\tau} \right).$$

- ▶ The Divisia growth rate appears as a benchmark:

$$g_{\tau}^D = s_e [s_{g,\tau} g_{c_g} + (1 - s_g) g_{c_s,\tau}] + (1 - s_e) g_x.$$

- ▶ Deviations from Divisia growth arise because fixed-base welfare comparisons imply different expenditure and sectoral shares.

$t \rightarrow$  preference date     $\tau \rightarrow$  welfare comparison date     $z \rightarrow$  reference price date

# Three Welfare-Based Indices

For any time  $z \in [t_0, t_1]$ , with reference time  $t_0$  and current time  $t_1$ .

## 1. Chained Fisher–Shell = Chained Divisia index

$$\mathcal{D}_z = \int_{t_0}^z g_h^D dh.$$

Local measure: we compare  $\tau$  with  $\tau - \Delta\tau$ , adopting the preferences and prices of  $\tau$

## 2. Current-base Fisher–Shell index

$$\mathcal{P}_{t_1,z} = \log \hat{m}_{t_1,t_1,z} - \log \hat{m}_{t_1,t_1,t_0}.$$

The past is evaluated using current preferences and prices (like a Paasche Index).

## 3. Reference-base Fisher–Shell index

$$\mathcal{L}_{t_0,z} = \log \hat{m}_{t_0,t_0,z} - \log m_{t_0}.$$

The future is evaluated using reference-time preferences and prices (Laspeyres Index).

# Proposition 5

## Current-Base and Reference-Base Fisher–Shell Indices

- At the base date, fixed-base Fisher–Shell growth coincides with Divisia growth:

$$\left. \frac{d\mathcal{P}_{t_1,z}}{dz} \right|_{z=t_1} = g_{t_1}^D, \quad \left. \frac{d\mathcal{L}_{t_0,z}}{dz} \right|_{z=t_0} = g_{t_0}^D.$$

- Away from the base date, the indices diverge:

$$\frac{d\mathcal{P}_{t_1,z}}{dz} < g_z^D < \frac{d\mathcal{L}_{t_0,z}}{dz} \quad \Rightarrow \quad \mathcal{P}_{t_1,z} < \mathcal{D}_z < \mathcal{L}_{t_0,z}.$$

**Current-base measures understate past growth.**

**Reference-base measures overstate it.**

$t$  → preference date     $\tau$  → welfare comparison date     $z$  → reference price date

# Plan

Introduction

Stylized Facts

Economic Environment

Economic Theory of Index Numbers

**Quantitative Implementation**

Main Quantitative Results

Conclusion

# Calibration

## Externally Calibrated Parameters

| $\theta$ | $\rho$ | $\delta$ | $n$    | $g_h$  | $g_A$  | $g_s$  | $g_g$  | $\zeta$ |
|----------|--------|----------|--------|--------|--------|--------|--------|---------|
| 0.333    | 0.040  | 0.080    | 0.0091 | 0.0046 | 0.0130 | 0.0024 | 0.0097 | 0.0137  |

- ▶  $\theta, \rho, \delta$  taken from HRV.
- ▶ Extend HRV data through 2023.
- ▶ Compute exponential growth rates of:
  - ▶ population ( $n$ ),
  - ▶ aggregate labor services ( $g_h$ ),
  - ▶ sectoral ( $\{g_s, g_g\}$ ) and aggregate ( $g_A$ ) TFP, via growth accounting.
- ▶  $\zeta$  matches the gap between observed and the model-implied relative prices.

# Calibration

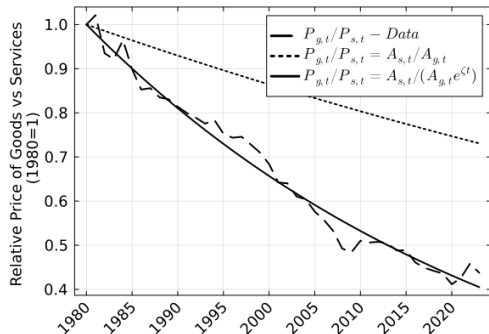
Preference Parameters Estimated via SMM

| $\chi$ | $\eta$ | $\gamma$ |
|--------|--------|----------|
| 0.360  | 0.251  | 0.714    |

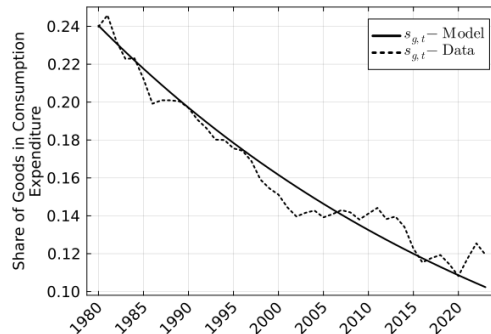
- ▶  $\eta$ ,  $\chi$ , and  $\gamma$  are estimated by matching the goods' share in consumption expenditure, 1980–2023.
- ▶ The preference system captures both income effects and relative-price effects in consumption.

# Calibration Fit

## Relative Prices and Expenditure Shares



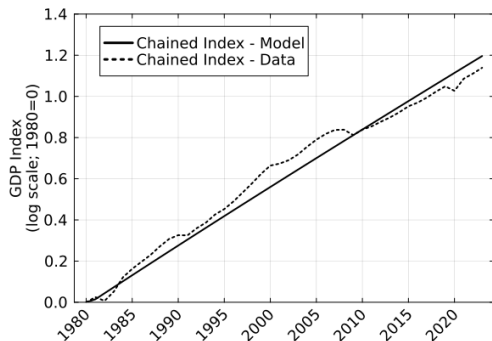
(a) Consumption goods relative price



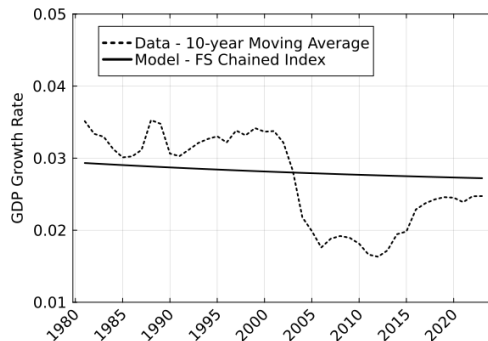
(b) Goods consumption share

# Model Fit

## GDP Level and Growth



(a) U.S. vs model GDP



(b) U.S. vs model GDP growth

# Plan

Introduction

Stylized Facts

Economic Environment

Economic Theory of Index Numbers

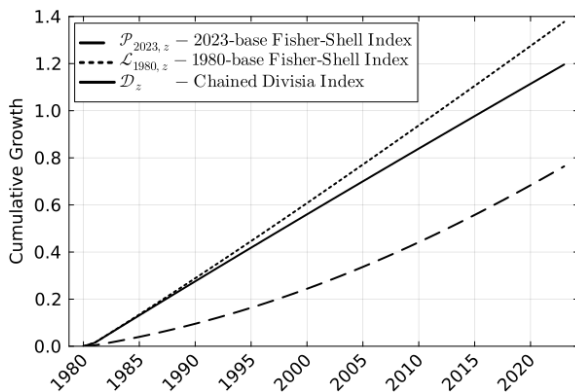
Quantitative Implementation

**Main Quantitative Results**

Conclusion

# Welfare-Based Real GDP Measures

## Fixed-Base Fisher-Shell Indices vs Chained Divisia Index

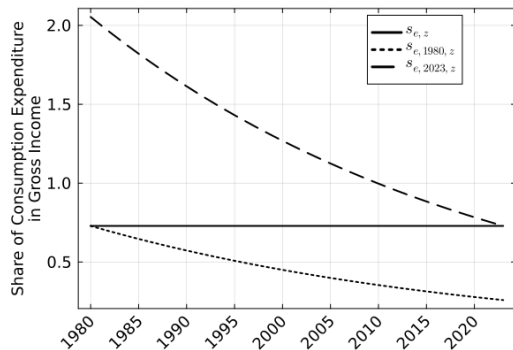


- ▶ The 1980-base Fisher–Shell index lies above the chained Divisia index (+18 p.p. in 2023).
- ▶ The 2023-base Fisher–Shell index lies below it (-43 p.p. in 2023).

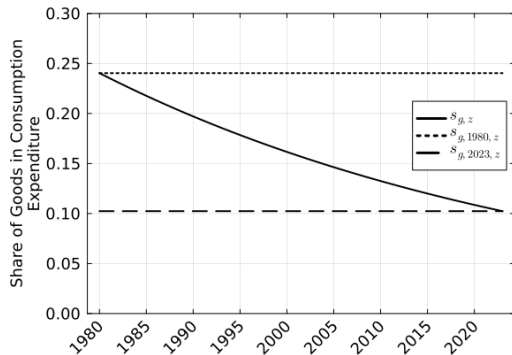
◀ Price-chained version

# Why Do the Indices Diverge?

## Implied Expenditure Shares



(a) Consumption expenditure share



(b) Goods consumption share

- Fixed-base EV measures evaluate welfare using expenditure shares that differ from the observed equilibrium allocation.

# Economic Interpretation

## Base-Year Bias

### ▶ 1980-base Fisher–Shell index

- ▶ uses past preferences to evaluate later allocations;
- ▶ assigns a declining weight to consumption;
- ▶ therefore overweights investment, the faster-growing component;
- ▶ overstates real income growth.

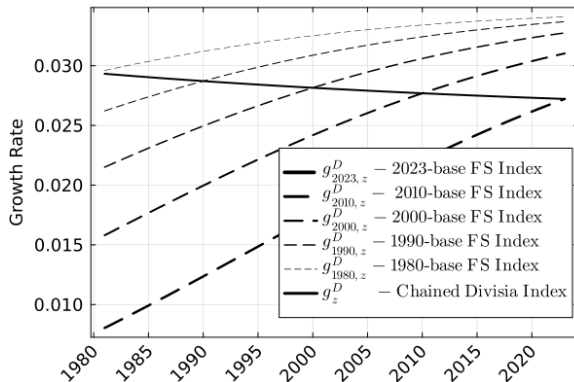
### ▶ 2023-base Fisher–Shell index

- ▶ uses current preferences to evaluate past allocations;
- ▶ assigns an excessively high weight to consumption in the past;
- ▶ therefore overweights the slower-growing component;
- ▶ understates real income growth.

**The chained Divisia index avoids these biases by updating prices and preferences locally.**

# Current-Time Upgrades

## GDP History Revisions



- Repeated current-base evaluations systematically revise past growth downward.

# Plan

Introduction

Stylized Facts

Economic Environment

Economic Theory of Index Numbers

Quantitative Implementation

Main Quantitative Results

Conclusion

# Conclusion

- ▶ Structural transformation and investment-specific technical change:
  - ▶ alter relative prices,
  - ▶ expenditure shares,
  - ▶ and the composition of output.
- ▶ Under these conditions:
  - ▶ welfare comparisons depend on the choice of preference base,
  - ▶ making fixed-base quantity indices systematically biased.
- ▶ The chained Divisia index:
  - ▶ continuously updates prices and expenditure weights,
  - ▶ coincides with the chained Fisher–Shell index in continuous time,
  - ▶ and provides a welfare-consistent measure of real income growth.
- ▶ Quantitatively:
  - ▶ fixed-base measures imply large and growing distortions,
  - ▶ and generate systematic revisions of historical growth.

# Thank You



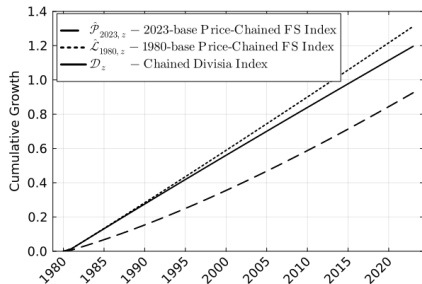
Replication files and online appendix available at:

<https://github.com/jivizcaino/Chained-Indices-Unchained>

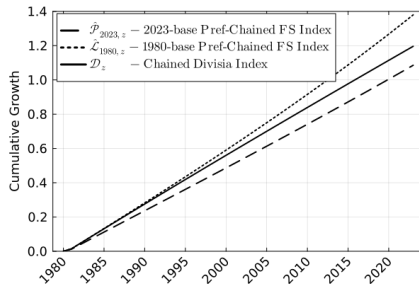
# Appendix

# Appendix

## Price Chained and Preference Chained Fisher–Shell Indices



(a) Price-chained index



(b) Preference-chained index

- (a) Prices update continuously, but preferences are kept fixed.
- (b) Preferences update continuously, but prices are kept fixed

**Most of the discrepancy with the Divisia index comes from holding preferences fixed.**

# Appendix

## Indices for Real Consumption Expenditure

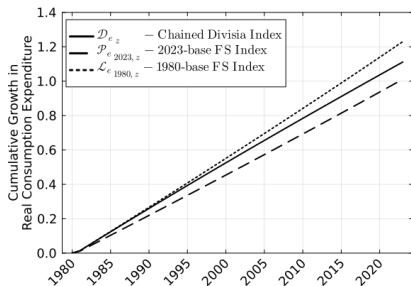
- In the endowment-economy analogue, the fixed-base Fisher–Shell growth rate for real consumption expenditure is

$$g_{e,t,z} = \frac{d \log \hat{e}_{t,z}}{dz} = g_{e,z}^D \left( \frac{e_z}{\hat{e}_{t,z}} \frac{P_{s,t}}{P_{s,z}} \right)^\chi.$$

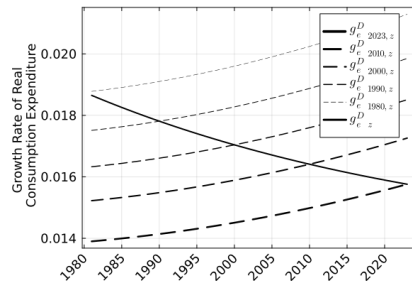
- A chained Fisher–Shell index again coincides with the chained Divisia index.
- Fixed-base indices still produce systematic base-year distortions under non-homotheticity.

# Appendix

## Real Consumption Expenditure Indices and Revisions



(a) Real consumption expenditure



(b) Growth-rate revisions

(a) Consumption expenditure explains part of the bias.

(b) Revisions are needed and fixed-based growth rates are increasing.

**Most of the discrepancy with the Divisia index comes from the consumption-investment trade-off.**

[◀ Back to main result](#)

# Appendix

## Transformation Distortions in Consumption

- ▶ Household expenditure prices differ from producer prices:

$$P_{j,t}^{\tau} = e^{\tau_j t} P_{j,t}, \quad j \in \{g, s\}.$$

- ▶ Producer prices and investment decisions are unchanged:

$$\frac{P_{g,t}}{P_{s,t}} = \frac{A_{s,t}}{A_{g,t}}.$$

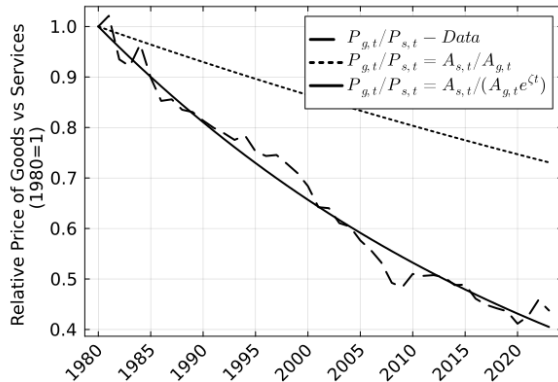
- ▶ Household relative expenditure prices become

$$\frac{P_{g,t}^{\tau}}{P_{s,t}^{\tau}} = \frac{A_{s,t}}{A_{g,t}} e^{(\tau_g - \tau_s)t}.$$

**The wedge affects consumption allocation, but not production decisions, factor prices, or the aggregate technology representation.**

# Appendix

## Transformation Wedge and Relative Prices



# Appendix

## Non-homothetic CES Preferences

- Replace PIGL demand with non-homothetic CES consumption:

$$\omega_c^{1/\sigma} \left( \frac{c_g}{C^{\eta_m}} \right)^{\frac{\sigma-1}{\sigma}} + (1 - \omega_c)^{1/\sigma} \left( \frac{c_s}{C^{\eta_s}} \right)^{\frac{\sigma-1}{\sigma}} = 1.$$

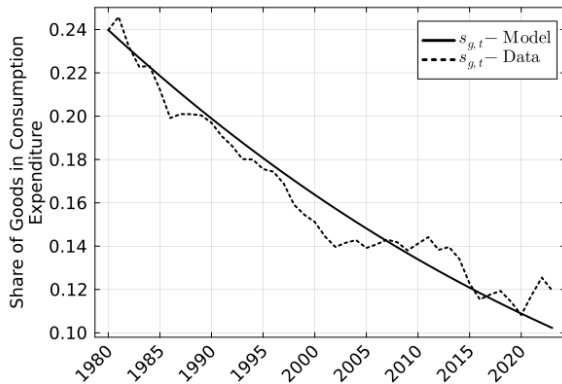
- The expenditure function is available in closed form:

$$E(P_g, P_s, C) = \left[ \omega_c P_g^{1-\sigma} C^{(1-\sigma)\eta_m} + (1 - \omega_c) P_s^{1-\sigma} C^{(1-\sigma)\eta_s} \right]^{\frac{1}{1-\sigma}}.$$

- The production side, investment block, factor prices, and aggregation result are unchanged.

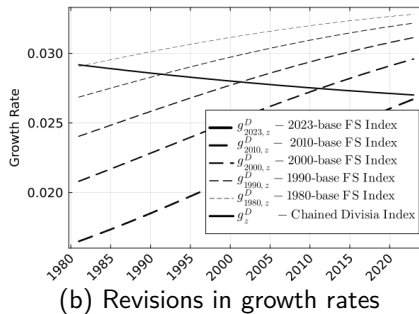
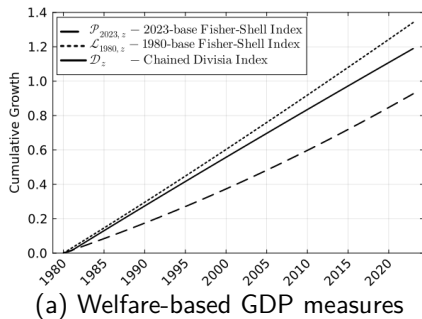
# Appendix

## NHCES Calibration Fit



# Appendix

## Welfare-based GDP measures and revisions under NHCES



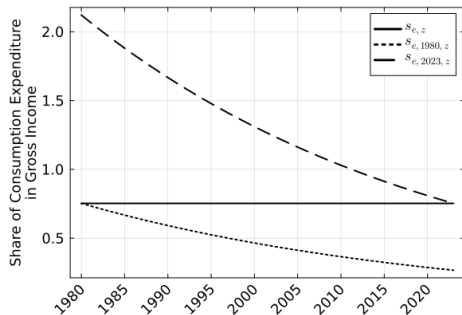
- (a) The same qualitative pattern holds: reference-base FS lies above Divisia; current-base FS lies below it.
- (b) Revisions are needed and fixed-base growth rates are increasing.

**Similar quantitative discrepancy with the Divisia under NHCES preferences.**

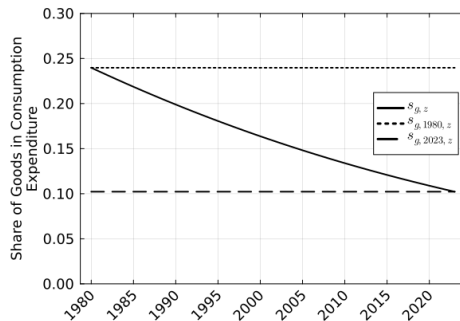
◀ Back to main result

# Appendix

Implied Expenditure Shares Under NHCES Preferences.



(a) Consumption expenditure share

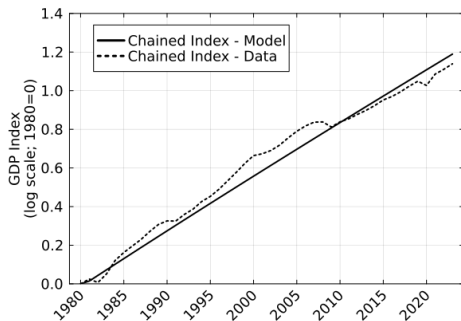


(b) Goods consumption share

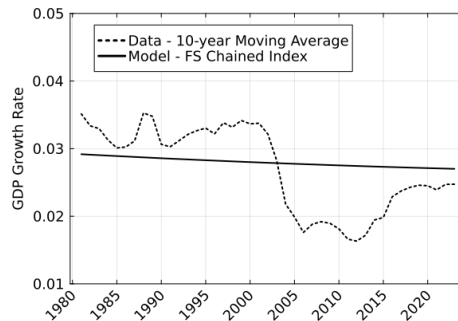
- Hypothetical expenditure shares are quantitatively similar under NHCES.

# Appendix

## Model fit Under NHCES



(a) GDP index



(b) GDP growth